

End of period t nominal government debt, denoted B_t , evolves as

$$B_t = G_t - T_t + DSC_t + R_t + B_{t-1} \quad (1)$$

where G_t = government spending, T_t = total tax revenues, DSC_t = debt service costs, and R_t = residual, as actual debt in t does not coincide with $G_t - T_t + DSC_t + B_{t-1}$, i.e.

$$R_t = B_t - (G_t - T_t + DSC_t + B_{t-1}).$$

The variable R_t is at the fund referred to as SFA_t .

Scaling equation (1) with lagged annualized nominal GDP $4Y_{t-1}$, we obtain

$$\frac{B_t}{4Y_{t-1}} = \frac{1}{4} \left(\frac{G_t}{Y_{t-1}} - \frac{T_t}{Y_{t-1}} + \frac{R_t}{Y_{t-1}} \right) + \frac{DSC_t}{B_{t-1}} \frac{B_{t-1}}{4Y_{t-1}} + \frac{B_{t-1}}{4Y_{t-1}}, \quad (2)$$

which can be rewritten as

$$\frac{B_t}{4Y_{t-1}} \frac{Y_t}{Y_t} = b_{y,t} y_t^{gr} = \frac{1}{4} (g_t - \tau_t + r_t) + \left(1 + i_t^{eff} \right) b_{y,t-1}, \quad (3)$$

where we have defined $b_{y,t} = \frac{B_t}{4Y_t}$, $y_t^{gr} = \frac{Y_t}{Y_{t-1}}$ and $i_t^{eff} = \frac{DSC_t}{B_{t-1}}$. We notice that i_t^{eff} is a quarterly interest rate, so to express it at an annualized rate it has to be multiplied by 400.

When the variables $g_t, \tau_t, r_t, i_t^{eff}$ and $b_{y,t-1}$ have been defined, we can use equation (2) to calculate $\frac{B_t}{4Y_{t-1}}$, and from equation (3) then we can simply invert

$$b_{y,t} = \frac{B_t}{4Y_{t-1}} \frac{1}{y_t^{gr}}, \quad (4)$$

and get $b_{y,t}$ which is used in the next forward itation. Thus, equations (3) and (4) together make up a simple two equation system to update government debt as share of GDP $\{b_{y,t}\}_{t=1}^T$ as a satellite model to a VAR model, given paths for the endogenous variables $\left\{ g_t, \tau_t, r_t, i_t^{eff}, y_t^{gr} \right\}_{t=1}^T$ given an initial condition $b_{y,0}$.